ACHARYA INSTITUTE OF TECHNOLOGY

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Department of Artificial Intelligence & Machine Learning
and
Computer Science & Engineering (Data Science)

PROJECT REPORT
ON

Big Data Analytics in Flipkart Minutes

Subject Name: Big Data Analytics
Subject Code: BDA601

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1. ABOUT THE PROJECT:

1.1 Background of Flipkart Minutes Data

Modern quick-commerce platforms like Flipkart Minutes generate enormous amounts of data every minute.

Every customer interaction produces digital records such as:

- Customer registration details
- Order transactions
- Product search logs
- Cart activity
- Payment transactions
- Delivery partner GPS tracking
- Warehouse inventory updates
- Customer reviews and ratings

In large cities:

- Thousands of orders are placed every hour
- Millions of product views are recorded daily
- Real-time delivery tracking generates streaming GPS data
- Product images and reviews create unstructured data

Traditional systems struggle to manage such high-volume, high-speed data efficiently.

This is where **Big Data Analytics in Flipkart Minutes** becomes essential.

1.2 Big Data in Flipkart Minutes – The 5V Model

Volume

- Order history
- Product catalog data
- Customer profiles
- Inventory logs
- Delivery records

Velocity
• Real-time order placement
• Instant payment confirmation
• Live delivery tracking
• Dynamic inventory updates

Variety
• Structured data (order ID, price, quantity)
• Semi-structured data (search logs, tracking updates)
• Unstructured data (reviews, images, feedback)

Veracity
Data may contain:
• Fake reviews
• Duplicate customer entries
• Incorrect delivery updates

Value
By analyzing Flipkart Minutes data, management can:
• Predict demand spikes
• Reduce delivery time
• Optimize warehouse stock
• Increase customer retention
• Improve profit margins

1.3 Why Traditional BI Fails in Flipkart Minutes

Traditional BI tools like Excel and simple SQL databases have limitations:

- Cannot process real-time delivery tracking.
- Cannot handle unstructured review data.
- Limited scalability.
- Slow with millions of transactions.
- Limited predictive capabilities.

Flipkart Minutes requires:

- Real-time analytics
- Distributed storage
- Demand forecasting
- Scalable infrastructure

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2. ABOUT TOOLS & TECHNOLOGIES (DETAILED)

2.1 Power BI – Analytics & Visualization Layer

Used for:
- Order performance dashboards
- Revenue analysis
- Area-wise sales tracking
- Delivery performance metrics
- Customer trend analysis

2.2 Apache Hadoop – Distributed Storage

Key Components:
• HDFS
• MapReduce
• YARN

In Flipkart Minutes:
• Stores massive transaction logs
• Stores customer browsing data
• Provides scalability and fault tolerance

2.3 Apache Spark – Real-Time Processing

Used for:
• Real-time order processing
• Demand forecasting
• Recommendation systems
• Fraud detection
Spark enables fast, in-memory processing.

2.4 MongoDB – NoSQL Database

Advantages:
• Flexible product schema
• Stores semi-structured data
• Handles dynamic customer behavior
• High availability

3. DETAILED DESCRIPTION OF CONTRIBUTION

3.1 What is Done and How it is Done

3.1.1 Data Understanding

Dataset represents Flipkart Minutes transactions.
Each row includes:
• order_id
• order_date
• customer_id
• product_id
• quantity
• delivery_status
• city

This dataset helps analyze:
• Order volume
• Customer behavior
• Delivery efficiency
• Sales trends

3.1.2 Data Preprocessing

Data Preprocessing

• Imported dataset into Power BI
• Verified data types
• Converted order_date into Date format
• Checked null values
• Created Month column for trend analysis
• Sorted data chronologically

3.1.3 Analytical Logic Applied

Dashboard focuses on:
- Total Orders
- Total Customers
- Total Revenue
- Completed Deliveries
- Cancelled Orders
- Monthly Order Trends
- Area-wise Demand

This helps management:
- Detect peak demand
- Monitor delivery efficiency
- Plan warehouse restocking
- Improve customer satisfaction

4. IMPLEMENTATION OF CODE

4.1 Total Orders

Total Orders = COUNT(orders[order_id])

4.2 Total Customers

Total Customers = DISTINCTCOUNT(orders[customer_id])

4.3 Total Revenue

Total Revenue = SUM(orders[price])

4.4 Completed Orders

Completed Orders =
CALCULATE(
COUNT(orders[order_id]),
orders[delivery_status] = "Delivered")

5. RESULTS ANALYSIS

5.1 Order Overview

Shows total orders, customers, and revenue.

5.2 Monthly Trend Analysis

Identifies peak shopping seasons.

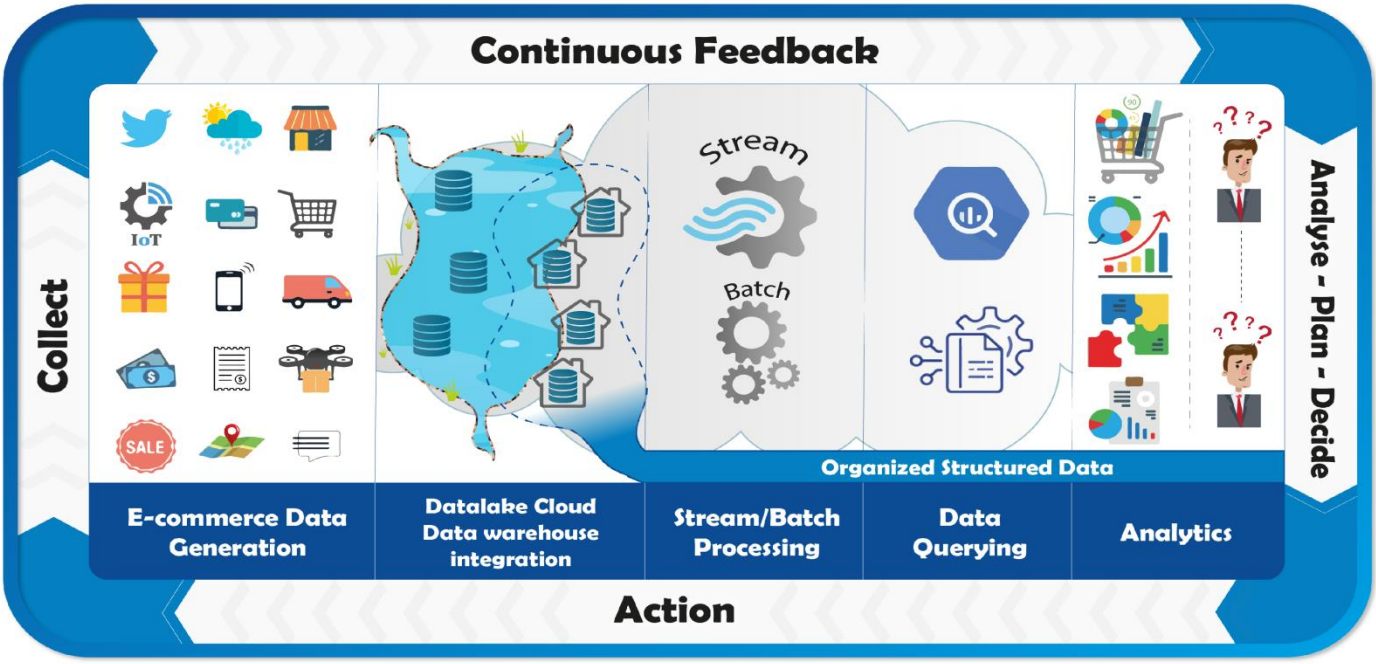
5.2 Delivery Performance

Highlights completed vs cancelled orders.

5.3 Area-wise Sales Insights

Identifies high-demand locations.

Figure 1.1: E-commerce Big Data Ecosystem



6. CONCLUSION:

This project demonstrates how Big Data Analytics improves quick-commerce platforms like Flipkart Minutes.

By integrating distributed storage, real-time processing, and interactive dashboards, Flipkart Minutes can:

- Deliver products faster
- Predict customer demand
- Reduce operational costs
- Improve customer satisfaction

Big Data transforms Flipkart Minutes from a simple shopping platform into an intelligent real-time commerce system.

Figure 1.2: Flipkart Minutes Dashboard created using Power BI.

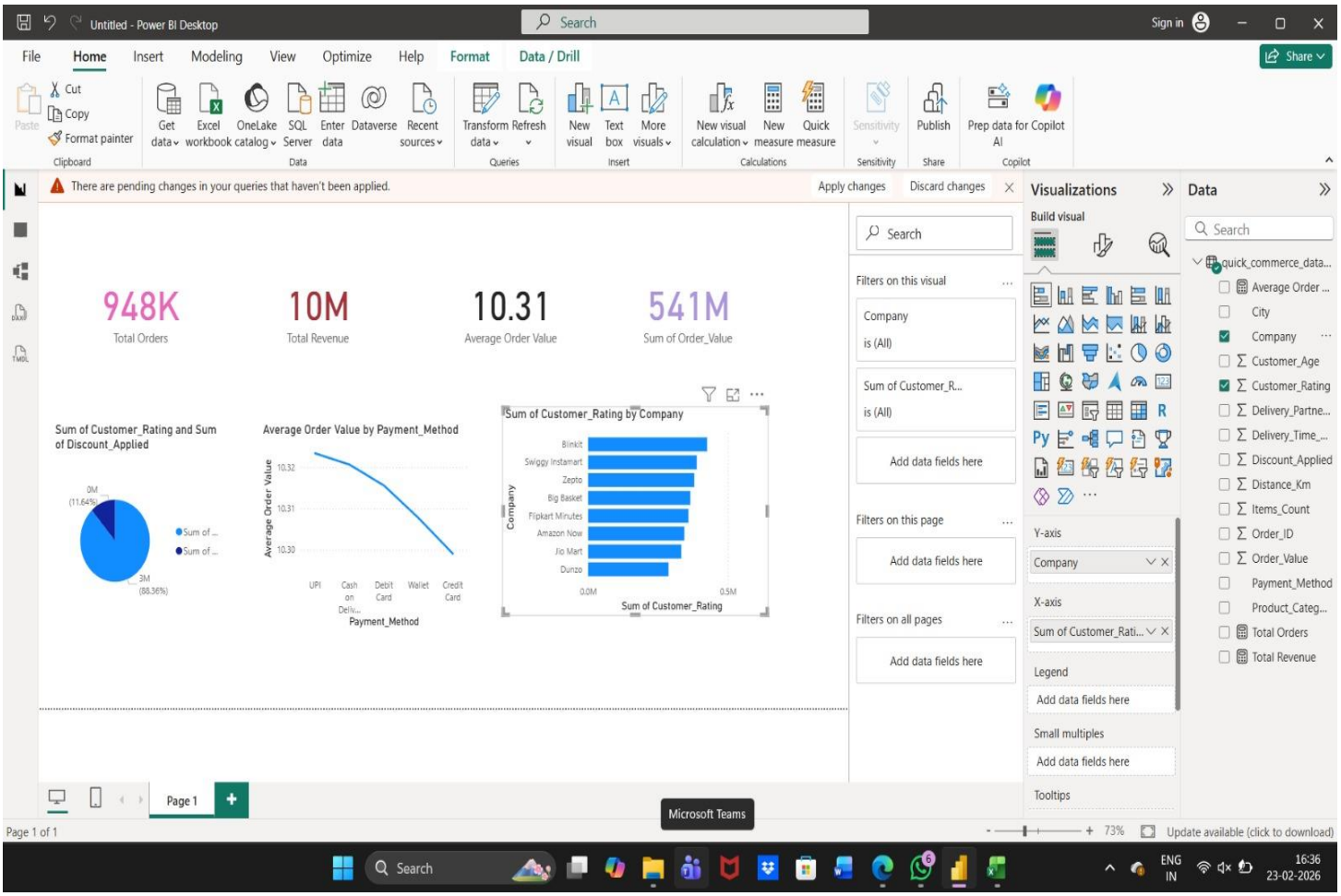
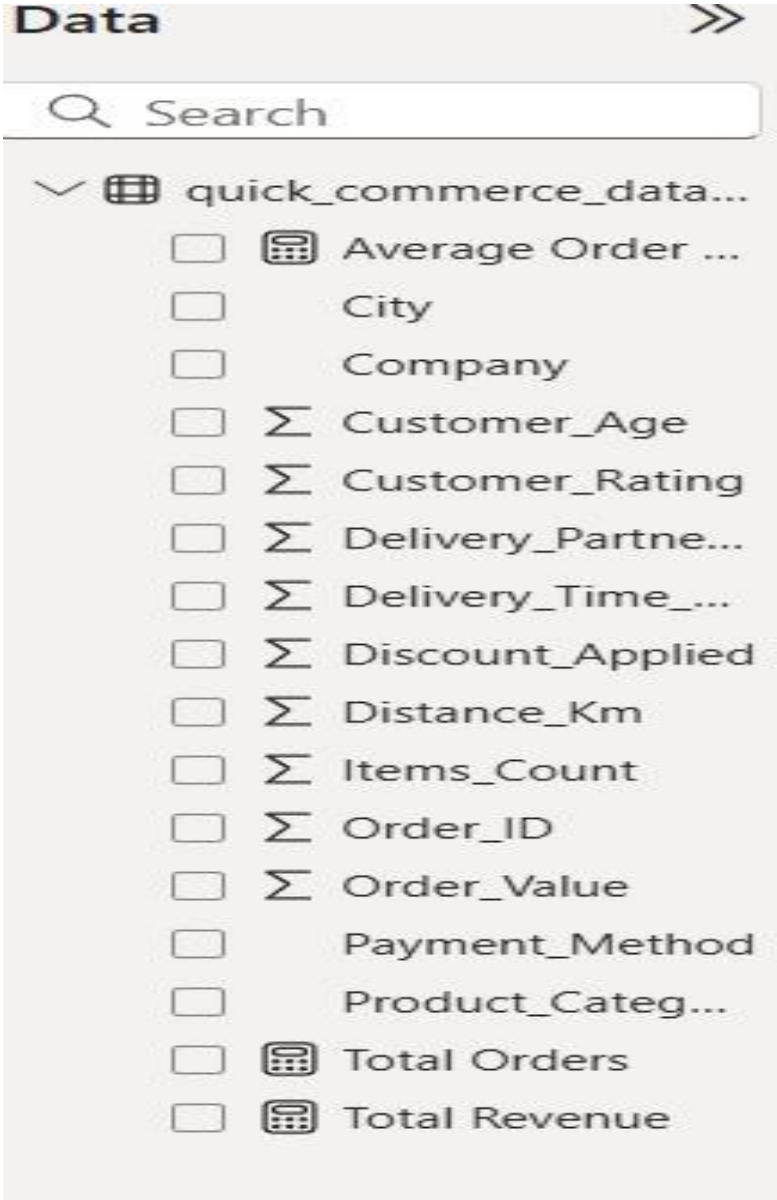


Figure 1.3:


Data
Search
quick_commerce_data...
Average Order ...
City
Company
Customer_Age
Customer_Rating
Delivery_Partne...
Delivery_Time_...
Discount_Applied
Distance_Km
Items_Count
Order_ID
Order_Value
Payment_Method
Product_Categ...
Total Orders
Total Revenue

TASK 2: Role Play – Big Data in Flipkart Minutes

Role: Data Analyst at Flipkart Minutes

Scenario:

Flipkart Minutes wants to reduce delivery time and increase customer satisfaction in metro cities.

Business Problems:

- Delayed deliveries in peak hours
- Sudden stock shortages
- Cart abandonment
- Low repeat customer rate

Data Available:

- Order history
- Customer browsing data
- GPS tracking data
- Inventory logs
- Payment records

My Role as Data Analyst:

- Analyze order trends by time and location
- Identify high-demand products
- Detect areas with delayed deliveries
- Study customer buying patterns
- Create dashboards for management

Suggested Solutions:

- Increase stock in high-demand dark stores
- Assign nearest delivery partner using GPS analytics
- Offer discounts during low-demand hours
- Improve route optimization

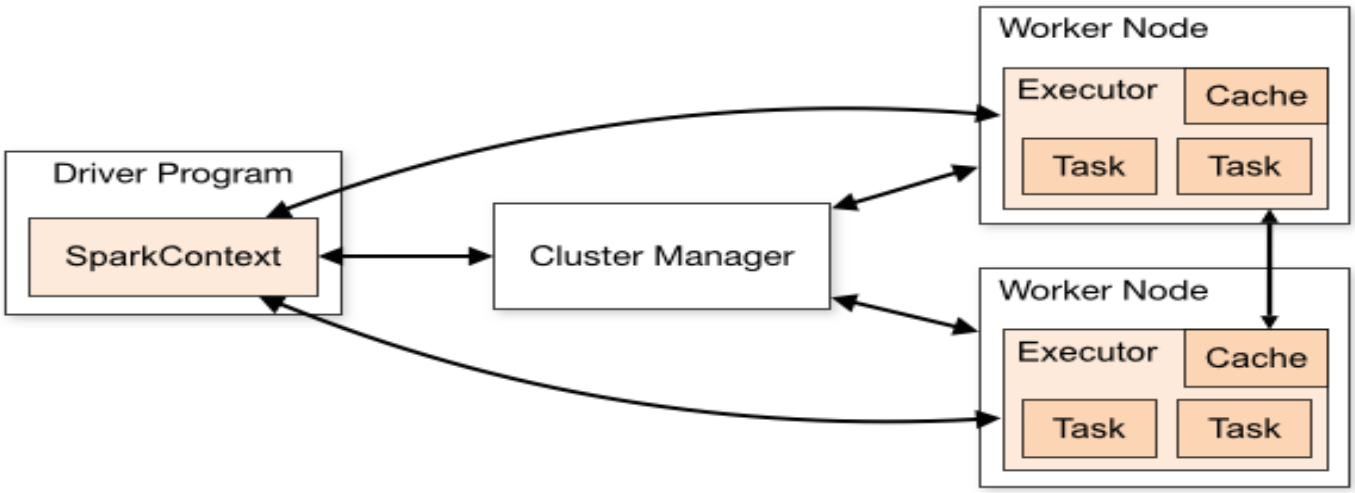
Expected Outcome:

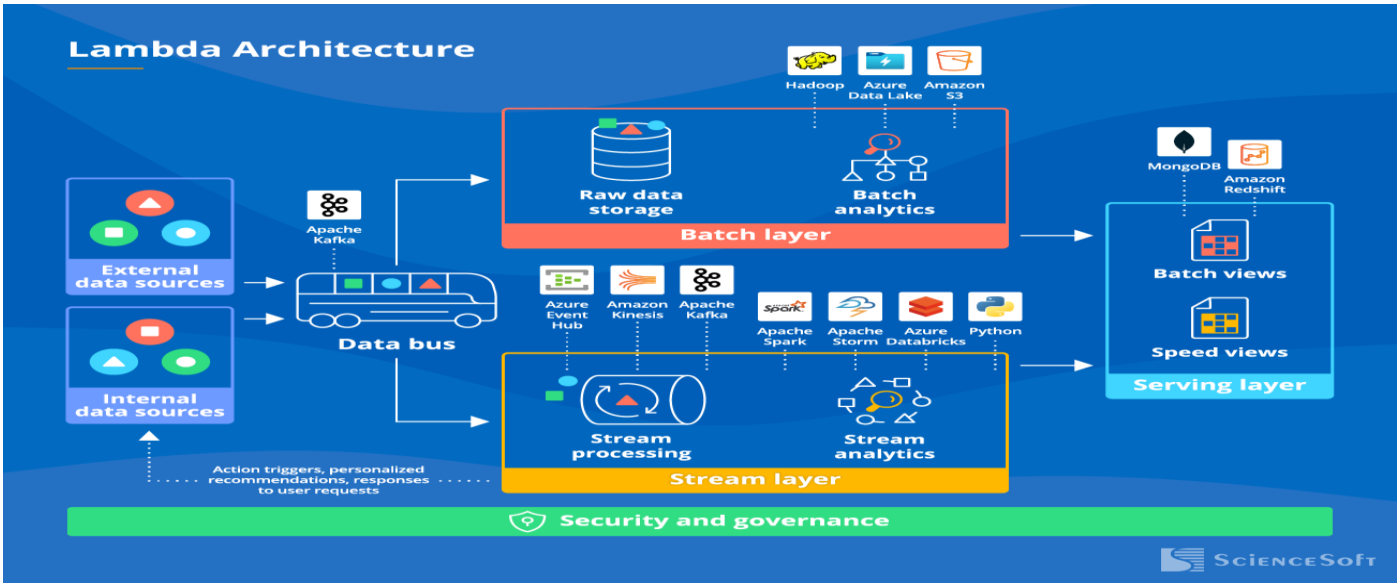
- Faster delivery
- Increased sales
- Improved customer retention
- Reduced operational cost

TASK 3: Big Data Architecture – Flipkart Minutes

Flipkart Minutes requires a scalable Big Data Architecture to process real-time orders.

Figure 3.1: Big Data Architecture for Quick Commerce





Architecture Layers:

1. **Data Sources Layer**
   - Customer App
   - Delivery Partner App
   - Warehouse System
   - Payment Gateway
2. **Data Ingestion Layer**
   - Apache Kafka (real-time streaming)
   - APIs
3. **Storage Layer**
   - HDFS (Hadoop Distributed File System)
   - NoSQL Databases
4. **Processing Layer**
   - Apache Spark (real-time analytics)
   - MapReduce
5. **Analytics & Visualization Layer**
   - Power BI
   - Dashboards
   - Reports

Working Flow:

Customer places order →
Data sent to Kafka →
Stored in HDFS →
Processed by Spark →
Dashboard updated in real-time →
Decision taken automatically

TASK 4: Analytics & Tool Match

Business Question	Analytics Type	Tool Used
What happened yesterday?	Descriptive	Power BI
Why did delivery delay increase?	Diagnostic	Spark
What will be demand tomorrow?	Predictive	ML Models
What action should be taken?	Prescriptive	Real-time Engine

BONUS CHALLENGE

1. AI-Based Demand Forecasting

Using:

- Historical order data
- Weather data
- Festival & weekend trends
- Area-wise purchase patterns

The system predicts:

- Which product will sell more
- In which location
- At what time

2. Real-Time Smart Delivery Routing

Using:

- GPS data
- Traffic data
- Order density
- Delivery partner availability

The system automatically:

- Assigns nearest delivery partner
- Optimizes route
- Reduces delivery time